energy-dispersive measurement (the sample had to be translated 50 mm to reach the monochromatic beam) are not smooth (*10*) (fig. S6). Instead they reveal grain growth, which is stronger in the β-brass than in the α-brass.

The experiment revealed that void growth kinetics during creep could be studied in the bulk material in situ. The access to 3D bulk information further provided access to a quantitative determination of void size and void shape evolution during creep. This comprehensive 3D representation of creep damage in the sample confirms the proposition that the transition from homogeneous to localized creep damage occurs late in creep life (*4*, *17*). In the system studied, the increase of pore volume versus time obeys an exponential growth law. The evolution of large pores coincides with texture formation in α-brass.

Diffraction and tomography performed in one single experiment should also make it possible to observe changes in the stress field and plastic zone around growing cracks, load partitioning and deformation in reinforced cellular and porous materials, the delamination in fiber-reinforced materials, and the resulting changes in strain respectively stress partitioning between the fibers and the matrix.

References and Notes

1. H. Riedel, *Fracture at High Temperatures* (Springer-Verlag, Berlin, 1987).
2. L. M. Kachanov, *Izv. Akad. Nauk SSSR Otd. Tekh. Nauk* **8**, 27 (1958).
3. Y. N. Rabotnov, *Creep Problems in Structural Members* (North Holland, Amsterdam, 1969).
4. G. Eggeler, *Acta Metall. Mater.* **39**, 221 (1991).
5. J.-Y. Buffière, E. Maire, P. Cloetens, G. Lormand, R. Fougères, *Acta Mater.* **47**, 1613 (1999).
6. M. Di Michiel *et al.*, *Rev. Sci. Inst.*, in press.
7. O. Lame, D. Bellet, M. Di Michiel, D. Bouvard, *Nucl. Instrum. Methods Phys. Res. B* **200**, 287 (2003).
8. W. Reimers *et al.*, *J. Mater. Sci. Lett.* **19**, 581 (1999).
9. R. Sinclair *et al.*, *Acta Mater.* **52**, 1423 (2004).
10. Materials and methods are available as supporting material on *Science* Online.
11. K. Mäder, E. Hornbogen, *Scripta Metall.* **8**, 979 (1974).
12. M. R. Willis, J. P. Jones, *Scripta Mater.* **44**, 31 (2001).
13. Information about the ESRF ID15A high-energy scattering beamline can be found at www.esrf.fr/UsersAndScience/Experiments/MaterialsScience/ID15.
14. A. C. Kak, M. Slaney, *Principles of Computerized Tomographic Imaging* (IEEE Press, New York, 1988), p. 186.
15. *IDL Reference Guide, Version 5.4,* (Research System Incorporated, Boulder, CO, 2000).
16. J. Grewen, G. Wassermann, *Z. Metallkde.* **45**, 498 (1954).
17. G. Eggeler, J. C. Earthman, N. Nilsvang, B. Ilschner, *Acta Metall.* **37**, 49 (1989).

18. The authors are grateful to H. M. Mayer for the tensile deformation of the samples before the experiment and D. Fernandez-Carreiras from the ESRF for the imaging detector control system. We acknowledge the ESRF for beamtime on ID15A and financial support of travelling expenses. A.P. and W.R. acknowledge the Deutsche Forschungsgemeinschaft (DFG) for financial support of project Py9/1-2.

Supporting Online Material

www.sciencemag.org/cgi/content/full/308/5718/92/DC1
Materials and Methods
Figs. S1 to S6
Movies S1 and S2

26 October 2004; accepted 10 February 2005
10.1126/science.1106778

U-Pb Ages from the Neoproterozoic Doushantuo Formation, China

Daniel Condon,[1]* Maoyan Zhu,[2] Samuel Bowring,[1] Wei Wang,[2] Aihua Yang,[2] Yugan Jin[2]

U-Pb zircon dates from volcanic ash beds within the Doushantuo Formation (China) indicate that its deposition occurred between 635 and 551 million years ago. The base records termination of the global-scale Marinoan glaciation and is coeval with similar dated rocks from Namibia, indicating synchronous deglaciation. Carbon isotopic and sequence-stratigraphic data imply that the spectacular animal fossils of the Doushantuo Formation are for the most part younger than 580 million years old. The uppermost Doushantuo Formation contains a pronounced negative carbonate carbon isotopic excursion, which we interpret as a global event at circa 551 million years ago.

The Neoproterozoic Doushantuo Formation (South China), deposited after a global-scale glaciation, contains fossil embryos, algae, achritarchs, and small bilaterians that are purportedly the Earth's earliest animals. This formation also contains a chemostratigraphic record of environmental change during its deposition with distinct $\delta^{13}C$ anomalies occurring at its base and top. Here, we report previously undetermined U-Pb zircon ages for volcanic ash beds from the lower and uppermost Doushantuo Formation (Yangtze Gorges area, South China), each of which is associated with globally correlated negative $\delta^{13}C$ anomalies.

In its type area along the Yangtze Gorges, the Doushantuo Formation comprises a succession of carbonate, shale, and phosphatic shale that is about 100 m thick. It can be correlated over 8000 km^2 to the south and west (*1*). The Doushantuo Formation is characterized by two transgressive sequences. The lower sequence begins above a distinctive Lower Dolomite Member (cap carbonate), which is overlain by black shale and phosphorite that grade upward into increasingly dolomitic micrite and grainstone. The upper sequence is characterized by a basal black shale and phosphorite unit that grades into dolomitic micrite. These two sequences are separated by an interbasinal sequence boundary, which in the more proximal Weng'an successions occurs as a karstic unconformity (*2*, *3*). The well-preserved animal embryos, algae, and small bilaterians of the Doushantuo Formation are from the phosphorites of the Upper Sequence (*2*, *3*) at Weng'an, Guizhou Province. The uppermost part of the Doushantuo Formation (Miaohe Member) is considered to be correlative to the lower part of a third transgressive cycle grading into carbonates of the Dengying Formation. Abundant macroscopic multicellular algae fossils known as the Miaohe biota are found in the black shale (*4*).

The Lower Dolomite Member overlies the Nantuo Tillite and is equated with other Marinoan-type cap carbonates worldwide because of its distinctive carbonate facies and associated negative $\delta^{13}C$ values (*5*). Fossils from the overlying Dengying Formation consist of large horizontal traces and the weakly mineralized metazoan *Cloudina* considered characteristic of the <10 million years (My) preceding the Ediacaran/Cambrian boundary (*6*). An ash bed below the Nantuo Formation (i.e., Datangpo Formation) has a U-Pb zircon age (±SEM) of 663 ± 4 million years ago (Ma) (*5*). The overlying Cambrian Meishucunian phosphorite unit (the Zhongyicun Member), which contains abundant phosphatized small shelly fossils, provides a minimum age of 538 ± 1.5 Ma (*7*). Barfod *et al.* (*8*) reported Lu-Hf and Pb-Pb whole-rock isochron ages on phosphorites from the lower part of the Upper Sequence of the Doushantuo Formation of 602 ± 48 Ma [mean square of weighted deviates (MSWD) = 2] and 599 ± 4 Ma (MSWD = 2.9), respectively. Chen *et al.* (*9*) reported Pb-Pb isochron dates of 598 ± 26 Ma (MSWD = 2.7) and 576 ± 14 Ma (MSWD = 0.4) from the lower and upper parts, respectively, of the same sequence.

[1]Department of Earth, Atmospheric, and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, MA 02139, USA. [2]Nanjing Institute of Geology and Paleontology, Chinese Academy of Sciences, Nanjing 210008, China.

*To whom correspondence should be addressed. E-mail: dcondon@mit.edu

Downloaded from https://www.science.org at Curtin University on May 27, 2026

We collected three volcanic ash samples from the Doushantuo Formation in the type sections of the Yangtze Gorges area. Sample YG-04-15 comes from a 1-cm-thick clay-rich ash bed 2.3 m above the base of the Doushantuo Formation (Wuhe-Gaojiaxi Section) within the Lower Dolomite Member. Zircons from this sample are variably discordant and define a linear array on a concordia diagram anchored by three concordant points (fig. S2). All points (n = 11) yield a weighted mean $^{207}Pb/^{206}Pb$ date of 635.4 ± 1.3 Ma (MSWD = 0.31), and the four concordant points yield a U-Pb "concordia age" (*10*) of 635.23 ± 0.57 Ma (MSWD of concordance and equivalence = 0.55) (*11*). The Lower Dolomite Member is the cap carbonate to the Nantuo Tillite (*5*, *12*–*14*). More specifically, ash bed YG-04-15 occurs at the transition from laminated limestone-dolostone (displaying sheet-crack and tepee-like structures) to laminated limestone-dolostone micrite [transition from C2 to C3 facies in the sense of (*13*)]. Most models for cap carbonate precipitation invoke high accumulation rates (*15*). Therefore, we consider the age of sample YG-04-15 to date to the end of the Nantuo glaciation.

A second ash bed (YG-04-2) occurs 9.5 m above the base of the Doushantuo Formation (Jijiawan Section) and 5 m above the top of the Lower Dolomite Member (Nantuo Cap Carbonate). Zircons from this ash bed yield variably discordant data (n = 9) that define a linear array on a concordia diagram anchored by three concordant data points (fig. S2). The U-Pb concordia age of 632.50 ± 0.48 Ma (MSWD of concordance and equivalence = 0.38) is indistinguishable from the weighted mean $^{207}Pb/^{206}Pb$ age of 632.4 ± 1.3 Ma (MSWD = 0.36). Comparing this age with YG-04-15 constrains the time interval between these two beds to between 1.7 and 3.8 My, and in the Jijiawan section this interval is represented by 3 m of dolomite overlain by about 5 m of black shale (*16*).

The age of the uppermost Doushantuo Formation (Miaohe Member) is constrained by the age of a third ash bed, JIN-04-2 (Jijiawan Section) that occurs at the top of the black shale member containing the Miaohe biota. Zircons from this ash bed yield variably discordant dates (n = 10) that define a linear array on a concordia diagram anchored by two concordant analyses (fig. S2). All data yield a weighted mean $^{207}Pb/^{206}Pb$ date of 550.55 ± 0.75 Ma (MSWD = 0.48). The two concordant analyses yield a U-Pb

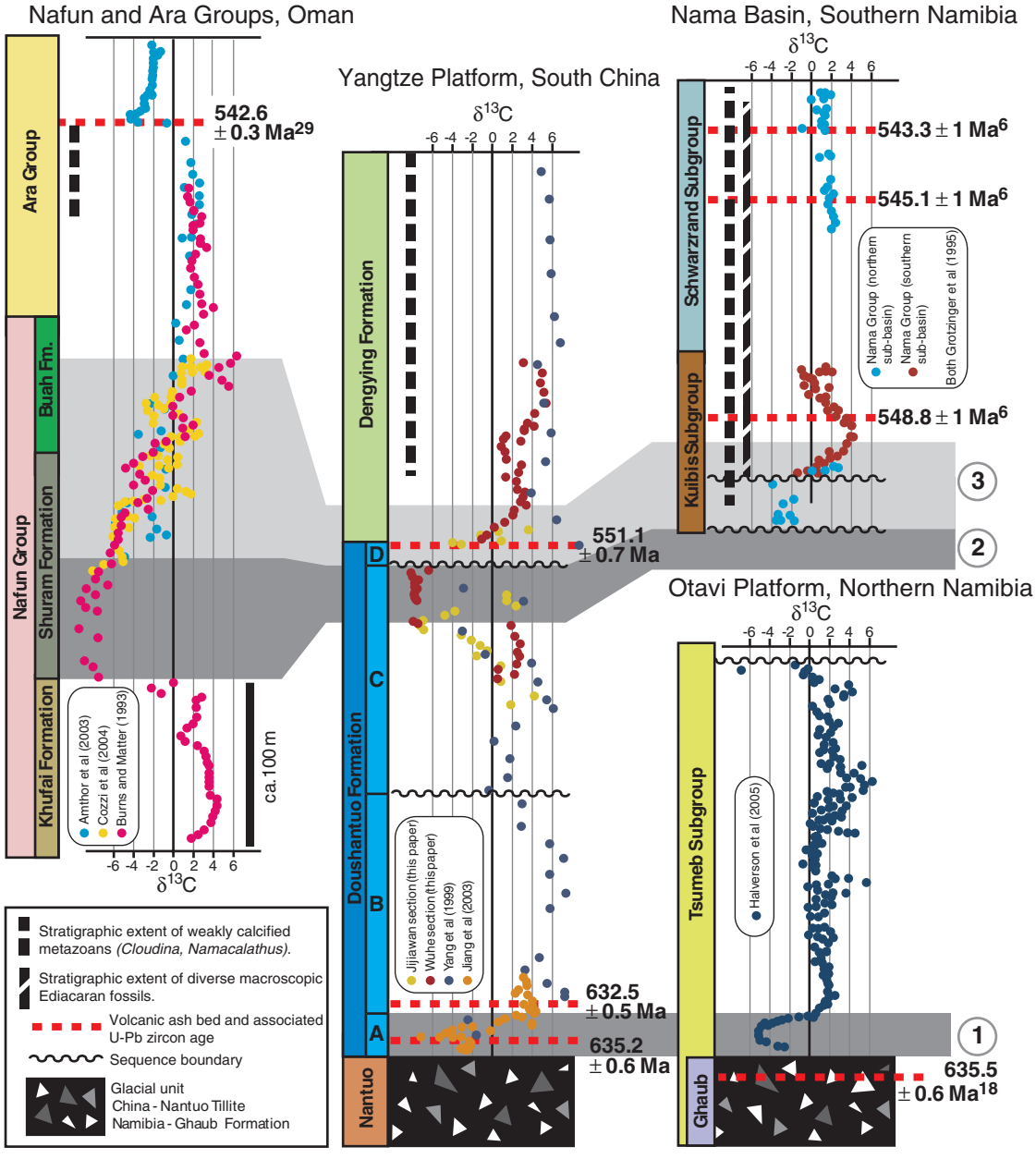


Fig. 1. Correlation of Doushantuo Formation (Yangtze Platform) with the Nama Group (Nama Basin) and Nafun/Ara Groups (Oman) successions. Three distinct sections are identified in each succession on the basis of magnitude of $\delta^{13}C$ and nature (increasing, decreasing, or invariant) of stratigraphic trend. (1) Correlation of cap carbonate excursion ($\delta^{13}C$ starting at –2‰, going to a nadir of –5‰, and returning to +2‰). (2) Invariant negative $\delta^{13}C$ excursion in the range of –7‰. (3) Transition from negative to positive (about 1‰) values after the upper Doushantuo/Shuram/Kuibis anomaly. Nama Group (Kuibis Formation) carbonates record values as low as –4‰. However, correlation with the Doushantuo Formation is based on stratigraphic proximity to the 549 Ma ash bed. Internal stratigraphy of the Doushantuo Formation: A, Lower Dolomite Member; B, Lower Sequence; C, Upper Sequence; and D, Miaohe Member. Superscripted numbers next to dates correspond to reference numbers.

Downloaded from https://www.science.org at Curtin University on May 27, 2026

concordia age of 551.07 ± 0.61 Ma (MSWD of concordance and equivalence = 0.48). This ash bed occurs about 85 cm below the base of the Dengying Formation (within the Miaohe Member) at the interface between a black shale unit and overlying carbonates that record a progressive increase in $\delta^{13}C$ values [from –4 to +0.5 per mil (‰)] over a thickness of 2.3 m. Negative $\delta^{13}C$ values occur both above and below the sequence boundary separating the Upper Sequence and Miaohe Member (fig. S1). Below this boundary a pronounced $\delta^{13}C$ excursion (with values as low as –8‰) occurs; in the Wuhe section it is characterized by invariant $\delta^{13}C$ values of –8‰, whereas in the Jijiawan section there is a return to positive $\delta^{13}C$ values before the base Miaohe Member sequence boundary is encountered. This local variation is most likely related to variable preservation below the sequence boundary and/or lateral variation in sediment accumulation and preservation. In both sections, the lowermost Dengying Formation dolomites have values of –3 to –1‰, which increase to about +3‰ over about 5 m. This trend is interpreted as being the top of the pronounced $\delta^{13}C$ excursion (with values as low as –8‰) that characterizes the top of the Doushantuo Formation (Fig. 1), thus constraining the age of the sustained negative excursion to (just) older than 551.1 ± 0.7 Ma, the time at which $\delta^{13}C$ values increase to positive values. This interpretation assumes that any period of nondeposition across this sequence boundary has a duration that is less than the duration of the negative $\delta^{13}C$ excursion itself.

The Doushantuo Formation contains two prominent $\delta^{13}C$ features, a negative (–5‰) to positive (up to 2‰) excursion in the Lower Dolomite Member (*5*, *13*) and a pronounced negative (down to –8‰) excursion in the uppermost Doushantuo Formation (Fig. 1 and fig. S1). The Nantuo Tillite–Lower Dolomite Member has been correlated with other Marinoan-type glacial successions recorded worldwide (*5*, *17*). Our age for the termination of the Marinoan glaciation is nearly identical to a concordant U-Pb age of 635.51 ± 0.54 Ma (*18*, *19*) marking the top of glacial deposits in the Ghaub Formation, Namibia. Termination of the Marinoan glaciation and cap carbonate deposition was thus essentially synchronous, within the age uncertainty, worldwide. These data support the hypothesis that cap carbonates were deposited very quickly after deglaciation (*20*).

It has been commonly assumed that the negative $\delta^{13}C$ anomaly at the top of the Doushantuo Formation was related to the Gaskiers glaciation (*21*, *22*), albeit with no sedimentological evidence for glaciation. If the lowermost Dengying Formation with its negative-to-positive $\delta^{13}C$ trend represents the top of this anomaly, this correlation is no longer viable because the 551.1 ± 0.7 Ma age from the uppermost Doushantuo ash bed (JIN-04-2, Jijiawan Section) postdates the circa 580 Ma Gaskiers glaciation. This age does reveal that much of the Upper Sequence and associated isotopic excursion occurred during a time when macroscopic Ediacaran Fauna are preserved in other successions worldwide [e.g., Mistaken Point Fauna (*23*); White Sea Fauna, and inferentially Australian Fauna (*24*)] and that their absence in the Doushantuo Formation is likely an artifact of paleoecological context (*25*).

Given that the top of the Doushantuo is not correlative with the Gaskiers glaciation, we suggest that the sequence boundary separating the Lower and Upper sequences (Fig. 1) within the Doushantuo reflects glacio-eustatic sea-level fluctuation associated with it. In the Yangtze Gorge region, there is no distinct evidence of a sea-level fluctuation. However, the more proximal succession at Weng'an, the corresponding surface, is karstic with evidence for subaerial exposure (*2*, *3*). An age of circa 580 Ma for this surface is at odds with the data from Barfod *et al.* (*8*), but given the complexity of diagenesis in phosphorites and the use of leached whole-rock samples that span several meters of section, it is difficult to evaluate the accuracy of the calculated dates. This interpretation would imply that the animal embryos and small bilaterian of the Weng'an Phosphorite Member (*2*, *3*) postdate the last extensive glaciation (circa 580 Ma) of the Neoproterozoic (*26*).

In Southern Namibia, carbonate rocks of the Schwarzrand and Kuibis subgroups (Southern and Northern Nama Basin) display a $\delta^{13}C$ excursion above a sequence boundary (*6*, *27*, *28*) and 500 m below an ash bed dated at 548.8 ± 1.0 Ma (*6*). Estimates of sediment accumulation rates in the Nama basin are about 100 to 200 m per 10^6 years, so the shift in $\delta^{13}C$ values is estimated to be about 551 Ma. In Oman, a distinct negative $\delta^{13}C$ anomaly in the lower half of the Shuram Formation (*29*–*31*) lies about 200 m below ash beds in the Ara Group dated at 542 Ma (*29*) and above the Fiq Formation, which is interpreted as a Marinoan-type glacial deposit (*32*). Given that in Namibia, Oman, and China, these similar sequences are overlain by rocks containing the Ediacaran/Cambrian boundary and their $\delta^{13}C$ curves display similar trends, we suggest correlation of the Shuram, Kuibis, and upper Doushantuo excursions. Halverson and colleagues (*17*) also correlated the Kuibis/Shuram anomalies with those of similar magnitude and stratigraphic position (i.e., post-Marinoan and pre-Cambrian) in Australia

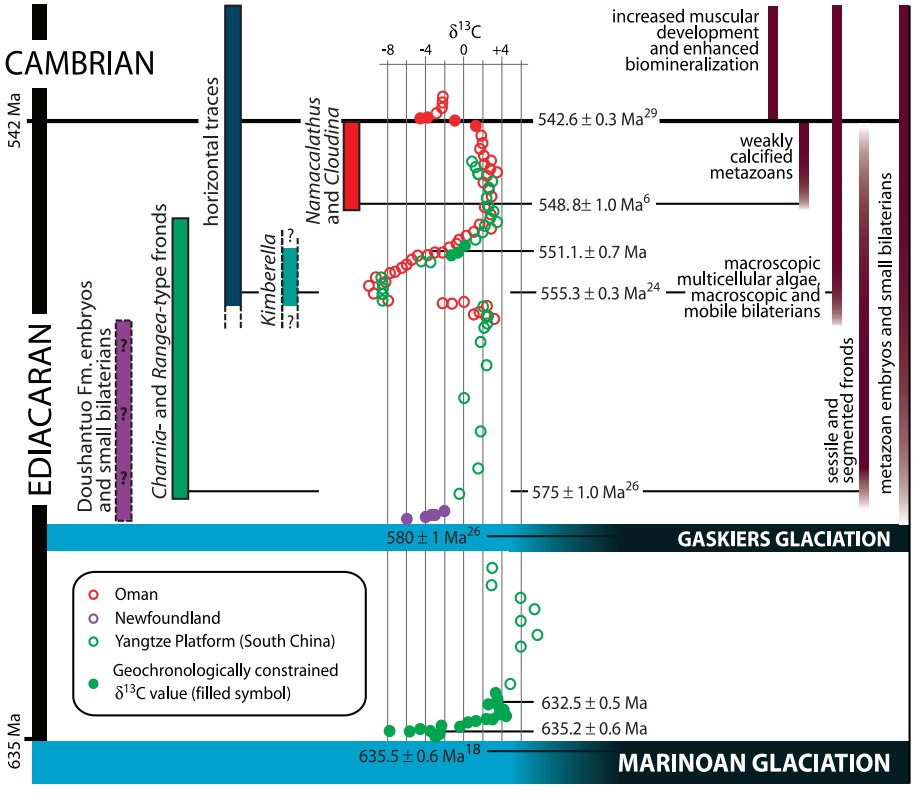


Fig. 2. Schematic diagram comparing fossil occurrences, evolutionary developments, and globally correlated $\delta^{13}C$ excursions within the context to a temporal framework constrained by U-Pb ages from sequences in Namibia, Oman, China, and Newfoundland. Newfoundland carbonate carbon $\delta^{13}C$ data are from Myrow and Kaufman (*35*). The absolute age of the –8‰ nadir is unconstrained except that it must be older than 551.1 ± 0.7 Ma. Superscripted numbers next to dates correspond to reference numbers.

Downloaded from https://www.science.org at Curtin University on May 27, 2026

[the Wonoka Formation (*33*)] and the western United States [Rainstorm Member, Johnnie Formation (*34*)] but chose to equate it with a postulated Gaskiers-related $\delta^{13}C$ excursion from circa 580 Ma (*35*). Instead, our data indicate that this globally correlated negative $\delta^{13}C$ excursion is not related to any known glaciation (*36*). The duration of this excursion is unconstrained; however, given that it is captured within over 100 m of section in Oman (Fig. 1) combined with approximate sediment accumulation rates calculated with our constraints, we suggest a duration of >1 and <10 My.

The Doushantuo and correlative strata record a fundamental shift from an interval of large carbon isotopic anomalies corresponding to glacial episodes (750 to 580 Ma) to an interval of anomalies unrelated to obvious glacial episodes (i.e., the anomalies from circa 551 and 542 Ma), as well as subsequent large fluctuations in the lower Cambrian. These new geochronological data allow us to calibrate that shift as being synchronous with the appearance of larger and more complex metazoans; this suggests possible feedback relationships between evolutionary innovation and seawater chemistry (Fig. 2).

Our ages indicate that the Doushantuo Formation spans more than 90% of the Ediacaran Period. These constraints are consistent with the upper Doushantuo/Shuram/Kuibis excursion being broadly coincident with the first appearance of complex trace fossils and mollusk-like bilaterian *Kimberella* (*37*), dated as slightly older than 555.1 ± 1.0 Ma (*24*). The advent of large pelagic bilaterians with unidirectional guts would have increased the flux of organic carbon to the deep ocean (*38*). Additionally, the radiation of algae containing resistant biopolymers in cell wall and cysts (i.e., Miaohe Biota) and the advent of biomineralization (*Namacalathus* and *Cloudina*, >549 Ma) would have also resulted in an increased organic carbon and carbonate carbon flux (*39*). These changes would have resulted in a downward flux of organic carbon with a possible coupled oxidation of the organic reservoir (*38*, *39*) driving the negative $\delta^{13}C$ excursion. This feedback loop would lead to an increase in marine oxygen levels and stimulate productivity and inferentially predation. It may be no coincidence that the first reefs inhabited by abundant weakly calcified and rare fully calcified metazoans appeared at about the same time as the isotopic anomaly [i.e., before 549 Ma in Namibia (*6*, *40*)].

References and Notes

1. M. Y. Zhu *et al.*, *Prog. Nat. Sci.* **13**, 951 (2003).
2. J. Y. Chen *et al.*, *Science* **305**, 218 (2004).
3. S. H. Xiao, Y. Zhang, A. H. Knoll, *Nature* **391**, 553 (1998).
4. S. H. Xiao, X. L. Yuan, M. Steiner, A. H. Knoll, *J. Paleontol.* **76**, 347 (2002).
5. C. M. Zhou *et al.*, *Geology* **32**, 437 (2004).
6. J. P. Grotzinger, S. A. Bowring, B. Z. Saylor, A. J. Kaufman, *Science* **270**, 598 (1995).
7. R. J. F. Jenkins, J. A. Cooper, W. Compston, *J. Geol. Soc. London* **159**, 645 (2002).
8. G. H. Barfod *et al.*, *Earth Planet. Sci. Lett.* **201**, 203 (2002).
9. D. F. Chen, W. Q. Dong, B. Q. Zhu, X. P. Chen, *Precambrian Res.* **132**, 123 (2004).
10. Materials and methods are available as supporting material on *Science* Online.
11. Age uncertainties are stated at the 2σ level.
12. J. D. Yang, W. G. Sun, Z. Z. Wang, Y. S. Xue, X. C. Tao, *Precambrian Res.* **93**, 215 (1999).
13. G. Jiang, M. J. Kennedy, N. Christie-Blick, *Nature* **426**, 822 (2003).
14. M. Macouin *et al.*, *Earth Planet. Sci. Lett.* **224**, 387 (2004).
15. P. F. Hoffman, D. P. Schrag, *Terra Nova* **14**, 129 (2002).
16. This corresponds to approximate sediment accumulation rates of about 1.5 to 3 m/My.
17. G. P. Halverson, P. F. Hoffman, D. P. Schrag, A. C. Maloof, A. H. N. Rice, *Geol. Soc. Am. Bull.*, in press.
18. K.-H. Hoffmann, D. J. Condon, S. A. Bowring, J. L. Crowley, *Geology* **32**, 817 (2004).
19. This is the analytical uncertainty on the U-Pb concordia age (excluding tracer calibration uncertainty) calculated from Hoffmann *et al.* (*18*).
20. P. F. Hoffman, A. J. Kaufman, G. P. Halverson, D. P. Schrag, *Science* **281**, 1342 (1998).
21. W. Wang *et al.*, *Acta Micropalaeontol. Sin.* **19**, 382 (2002).
22. S. H. Xiao *et al.*, *Precambrian Res.* **130**, 1 (2004).
23. G. M. Narbonne, J. G. Gehling, *Geology* **31**, 27 (2003).
24. M. W. Martin *et al.*, *Science* **288**, 841 (2000).
25. D. Grazhdankin, *Paleobiology* **30**, 203 (2004).
26. S. Bowring, P. Myrow, E. Landing, J. Ramezani, J. Grotzinger, *Geophys. Res. Abstr.* **5**, 13219 (2003).
27. The Schwarzrand Subgroup is characterized by a +2 plateau (*6*,*28*). In the northern subbasin, the underlying Kuibis Subgroup records a decrease from +3 down to 0 in upper Mooifontein Member with values of −3 obtained on carbonates from the underlying Mara Member. In the southern subbasin the Kuibis Subgroup has values of about 0 at the top that gradually increase toward the bottom of the section to +4 and then decrease down to −1.
28. B. Z. Saylor, A. J. Kaufman, J. P. Grotzinger, F. Urban, *J. Sediment. Res.* **68**, 1223 (1998).
29. J. E. Amthor *et al.*, *Geology* **31**, 431 (2003).
30. S. J. Burns, A. Matter, *Eclogae Geol. Helv.* **86**, 595 (1993).
31. A. Cozzi, J. P. Grotzinger, P. A. Allen, *Geol. Soc. Am. Bull.* **116**, 1367 (2004).
32. P. A. Allen *et al.*, in *16th International Sedimentological Congress Abstract Volume* (International Association of Sedimentologists, Johannesburg, South Africa, 2002), pp. 7–8.
33. C. R. Calver, *Precambrian Res.* **100**, 121 (2000).
34. F. A. Corsetti, A. J. Kaufman, *Geol. Soc. Am. Bull.* **115**, 916 (2003).
35. P. M. Myrow, A. J. Kaufman, *J. Sediment. Res.* **69**, 784 (1999).
36. G. Q. Jiang, L. E. Sohl, N. Christie-Blick, *Geology* **31**, 917 (2003).
37. M. A. Fedonkin, B. M. Waggoner, *Nature* **388**, 868 (1997).
38. G. A. Logan, J. M. Hayes, G. B. Hieshima, R. E. Summons, *Nature* **376**, 53 (1995).
39. D. H. Rothman, J. M. Hayes, R. E. Summons, *Proc. Natl. Acad. Sci. U.S.A.* **100**, 8124 (2003).
40. R. A. Wood, J. P. Grotzinger, J. A. D. Dickson, *Science* **296**, 2383 (2002).

41. This work was supported by NASA's Astrobiology Program (NCC2-1053 to S.B.), National Natural Science Foundation of China, the Chinese Academy of Sciences (KZCX3-SW-141), and Ministry of Science and Technology of China (G2000077700 and 2002CCA03300). We thank J. Zhang and X. Chen for fieldwork and C-O isotope analyses, respectively, and D. Fike, J. Grotzinger, G. P. Halverson, P. Hoffman, A. Knoll, A. Maloof, T. Prave, R. Summons, and two anonymous reviewers for comments. We thank D. Rothman for comments and discussion.

Supporting Online Material

www.sciencemag.org/cgi/content/full/1107765/DC1
Materials and Methods
Figs. S1 and S2
Tables S1 and S2
References

22 November 2004; accepted 14 February 2005
Published online 24 February 2005;
10.1126/science.1107765
Include this information when citing this paper.

Downloaded from https://www.science.org at Curtin University on May 27, 2026

Mortality and Greenhouse Gas Impacts of Biomass and Petroleum Energy Futures in Africa

Robert Bailis,[1] Majid Ezzati,[2]* Daniel M. Kammen[1,3]*

We analyzed the mortality impacts and greenhouse gas (GHG) emissions produced by household energy use in Africa. Under a business-as-usual (BAU) scenario, household indoor air pollution will cause an estimated 9.8 million premature deaths by the year 2030. Gradual and rapid transitions to charcoal would delay 1.0 million and 2.8 million deaths, respectively; similar transitions to petroleum fuels would delay 1.3 million and 3.7 million deaths. Cumulative BAU GHG emissions will be 6.7 billion tons of carbon by 2050, which is 5.6% of Africa's total emissions. Large shifts to the use of fossil fuels would reduce GHG emissions by 1 to 10%. Charcoal-intensive future scenarios using current practices increase emissions by 140 to 190%; the increase can be reduced to 5 to 36% using currently available technologies for sustainable production or potentially reduced even more with investment in technological innovation.

Biomass fuels (wood, charcoal, dung, and agricultural residues) are vital to basic welfare and economic activity in developing nations, especially in sub-Saharan Africa (SSA), where they meet more than 90% of household energy needs in many nations. Combustion of biofuels emits pollutants that currently cause over 1.6 million annual deaths globally (400,000 in SSA) (*1*). Because most of these deaths are among children and women, biomass use is

Science
AAAS

U-Pb Ages from the Neoproterozoic Doushantuo Formation, China

Daniel Condon, Maoyan Zhu, Samuel Bowring, Wei Wang, Aihua Yang, and Yugan Jin

Science **308** (5718), . DOI: 10.1126/science.1107765

View the article online
https://www.science.org/doi/10.1126/science.1107765
Permissions
https://www.science.org/help/reprints-and-permissions

Downloaded from https://www.science.org at Curtin University on May 27, 2026

Use of this article is subject to the Terms of service

Science (ISSN 1095-9203) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science* is a registered trademark of AAAS.

American Association for the Advancement of Science